

Electronics

Course Notes – Topic-wise Overview

Page 1: Basic Electronics

Topics covered:

- Voltage, current and resistance
- Ohm's and Kirchhoff's laws
- Semiconductor fundamentals
- Diodes and transistor basics

Learning Focus

This section introduces the important concepts of basic electronics. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 2: Analog & Digital Electronics

Topics covered:

- Amplifiers and filters
- Oscillators and power supplies
- Logic gates and Boolean algebra
- Flip-flops, counters and registers

Learning Focus

This section introduces the important concepts of analog & digital electronics. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 3: Embedded Systems

Topics covered:

- Microprocessors and microcontrollers
- Memory and I/O interfacing
- Sensors and actuators
- Embedded system design concepts

Learning Focus

This section introduces the important concepts of embedded systems. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.

Page 4: Communication & IoT

Topics covered:

- Communication system fundamentals
- Wireless communication
- Internet of Things
- PCB and circuit design basics

Learning Focus

This section introduces the important concepts of communication & iot. Students can use these topics as a starting point for lectures, assignments, revision and further study. Practical examples and laboratory activities can be added according to the college syllabus.